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Patent Public Search | Text View

United States Patent Application Publication

20250278220

Kind Code

A1

Publication Date

September 04, 2025

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INFORMATION PROCESSING APPARATUS AND NON-TRANSITORY COMPUTER-READABLE STORAGE MEDIUM STORING INFORMATION PROCESSING PROGRAM

Abstract

Provided is an information processing apparatus including: a color value acquisition unit that acquires a first color value indicating a color of a first color patch included in a color chart and a second color value indicating a color of a second color patch; a reproducibility acquisition unit that acquires a first reproducibility indicating color reproducibility when the first color patch is printed by a printing apparatus based on the first color value and acquires a second reproducibility indicating color reproducibility when the second color patch is printed by the printing apparatus based on the second color value; and a generation unit that generates print data for printing the color chart, in which when the first reproducibility is lower than the second reproducibility, the generation unit generates the print data for printing the color chart in which the first color patch and the second color patch have different representations.

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Family ID: 96881334

Appl. No.: 19/066239

Filed: February 28, 2025

Foreign Application Priority Data

JP 2024-031372

Mar. 01, 2024

Publication Classification

Int. Cl.: G06F3/12 (20060101)

Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-031372, filed Mar. 1, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an information processing apparatus and a non-transitory computer-readable storage medium storing an information processing program.

2. Related Art

[0003] Traditionally, a configuration is known in which a color chart including multiple patches is printed, and color adjustment is performed based on the color chart. For example, Japanese Patent No. 7314628 discloses a technique in which a center patch and multiple patches with altered Lab values of the center patch are printed on a color chart.

[0004] In the related art, although the usage amount of color materials can be specified by CMYK values, if the usage amount is excessive, only colors different from the intended colors may be printed.

SUMMARY

[0005] An information processing apparatus according to an aspect includes: a color value acquisition unit that acquires a first color value indicating a color of a first color patch included in a color chart and a second color value indicating a color of a second color patch included in the color chart; a reproducibility acquisition unit that acquires a first reproducibility indicating color reproducibility when the first color patch is printed by a printing apparatus based on the first color value and acquires a second reproducibility indicating color reproducibility when the second color patch is printed by the printing apparatus based on the second color value; and a generation unit that generates print data for printing the color chart, when the first reproducibility is lower than the second reproducibility, the generation unit generates the print data for printing the color chart in which the first color patch and the second color patch have different representations.

[0006] A non-transitory computer-readable storage medium storing an information processing program, according to another aspect, the program causing a computer to function as: a color value acquisition unit that acquires a first color value indicating a color of a first color patch included in a color chart and a second color value indicating a color of a second color patch included in the color chart; a reproducibility acquisition unit that acquires a first reproducibility indicating color reproducibility when the first color patch is printed by a printing apparatus based on the first color value and acquires a second reproducibility indicating color reproducibility when the second color patch is printed by the printing apparatus based on the second color value; and a generation unit that generates print data for printing the color chart, when the first reproducibility is lower than the second reproducibility, the generation unit generates the print data for printing the color chart in which the first color patch and the second color patch have different representations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a diagram illustrating a usage example of the information processing apparatus.

[0008] FIG. 2 is a block diagram of the colorimetric instrument.

[0009] FIG. 3 is a block diagram of the printing apparatus.

[0010] FIG. 4 is a block diagram of the information processing apparatus.

[0011] FIG. 5 is a flowchart of the print control process.

[0012] FIG. 6 is a diagram illustrating an example of the usage amount specification screen.

[0013] FIG. 7 is a diagram illustrating an example of the creation condition specification screen.

[0014] FIG. 8 is a diagram illustrating an arrangement example of color patches.

[0015] FIG. 9 is a diagram illustrating color values in a device-independent color space.

DESCRIPTION OF EMBODIMENTS

[0016] Here, embodiments of the present disclosure will be described below in the following order:

[0017] (1) System Configuration [0018] (1-1) Configuration of Colorimetric Instrument [0019] (1-2) Configuration of Printing Apparatus [0020] (1-3) Configuration of Information Processing Apparatus [0021] (2) Print Control Process [0022] (3) Other Embodiments, etc.

(1) System Configuration

[0023] FIG. 1 is a diagram illustrating a usage example of the information processing apparatus **10** according to an embodiment. In the present embodiment, the information processing apparatus **10** is connected to a colorimetric instrument **20** and a printing apparatus **30**. The information processing apparatus **10** is an apparatus that performs color conversion on image data to generate print data and causes the printing apparatus **30** to execute printing based on the print data. The user can also specify a desired color and print a color chart; and after confirming the color reproducibility based on the color chart, the user can execute printing of the desired image. In the present embodiment, the user can specify a reference color included in the color chart by measuring a color sample using the colorimetric instrument **20**. The color chart includes a patch printed based on a color value indicating the reference color and patches printed based on the color values of surrounding colors generated by varying the reference color.

(1-1) Configuration of Colorimetric Instrument

[0024] FIG. 2 is a block diagram illustrating the configuration of the colorimetric instrument **20**. The colorimetric instrument **20** includes a processor **20a**, a communication unit **20b**, non-volatile memory **20c**, a sensor **20d**, and a UI unit **20e**. The processor **20a** includes an unillustrated CPU, ROM, RAM, etc., executes a control program recorded in the non-volatile memory **20c**, and can control each unit of the colorimetric instrument **20**.

[0025] The processor **20a** may be configured with a single chip, multiple chips, or as an SoC with various functional blocks. For example, an ASIC may be employed instead of the CPU, or a configuration in which the CPU and ASIC cooperate may be employed. When each apparatus in the present embodiment includes a processor, the processor can be implemented in various ways, similar to the processor **20a**.

[0026] The communication unit **20b** includes a communication interface for communicating with the information processing apparatus **10** according to various wired or wireless communication protocols. The sensor **20d** is a device that irradiates a colorimetric target with light of a predetermined color temperature and detects the spectral distribution of reflected light. The processor **20a** acquires a color value (e.g., CIELAB value, HSV value, XYZ value, etc.) indicating the color of the colorimetric target by a color in a device-independent color space, based on the result read by the sensor **20d**. Hereinafter, the description will continue assuming that the color value is an HSV value (hue value, saturation value, and brightness value).

[0027] In the present embodiment, the user can specify the color value of the reference color based on the colorimetric result of the colorimetric instrument **20**. In the present embodiment, a sample color is measured by the colorimetric instrument **20** to specify the color value of the reference color. The sample color may be a color printed on various objects; however, here, it is assumed that a medium colored to be a sample color is measured. Thus, according to the configuration in which the reference color is specified by the colorimetric result of the colorimetric instrument **20**, the

reference color can be accurately specified.

[0028] In the present embodiment, when color matching is performed using the color chart, printing is performed so that the color value is reproduced for a specific single color; furthermore, the printing apparatus **30** can print colors similar to the color-matched color as colors reflecting the result of the color matching. Thus, in order to reflect the color matching result in multiple colors, in the present embodiment, the image data is composed of one or more layers.

[0029] In the present embodiment, each layer is called a plate. The number of color channels that can be used in each plate is predetermined; in the present embodiment, there is a process color plate that can be used by desirably varying any color over the entire range of gradation values, and a spot color plate for printing the color matched based on the color chart. The process color plate is a plate in which all usable colors can be desirably varied and used, and examples include image data in which all RGB (R: red, G: green, B: blue) can be used and image data in which all CMYK (C: cyan, M: magenta, Y: yellow, K: black) can be used.

[0030] Spot color plate image data consists of a representative color, which is the color representing the spot color plate, and colors with different densities from the representative color. In the spot color plate image data, the representative color is specified by each gradation value of RGB or CMYK in the header, and the color of each pixel is specified by a gradation value representing the relative density difference from the representative color. For example, when expressing the gradation value of a spot color plate with a 1-channel 8-bit gradation value, a gradation value of 128 matches the representative color, pixels with gradation values greater than 128 are considered darker than the representative color, and pixels with gradation values less than 128 are considered lighter than the representative color. A gradation value of 255 indicates the color when the representative color is made as dark as possible, and a gradation value of 0 indicates the color when the representative color is made as light as possible.

[0031] In the present embodiment, the representative color of the spot color plate is determined by the color matched using the color chart. That is, the user specifies the representative color by specifying one of the patches printed on the color chart. In the present embodiment, when a color sample is measured by the colorimetric instrument **20** to specify the reference color included in the color chart, the colorimetric data **20c1** indicating the color value, which is the colorimetric result, is stored in the non-volatile memory **20c**.

[0032] The UI unit **20e** is a button or the like provided on the housing of the colorimetric instrument **20**. The user can instruct the start of color sample measurement, etc., by operating the UI unit **20e**. The processor **20a** receives the user's instruction based on the output information of the UI unit **20e**.

(1-2) Configuration of Printing Apparatus

[0033] FIG. **3** is a block diagram illustrating the configuration of the printing apparatus **30**. The printing apparatus **30** includes a processor **30a**, a communication unit **30b**, non-volatile memory **30c**, a printing unit **30d**, and a UI unit **30e**. The processor **30a** includes an unillustrated CPU, ROM, RAM, etc., executes a control program recorded in the non-volatile memory **30c**, and can control each unit of the printing apparatus **30**.

[0034] The communication unit **30b** includes a communication interface for communicating with the information processing apparatus **10** according to various wired or wireless communication protocols. The communication unit **30b** may also include an interface for communicating with various removable memories attached to the printing apparatus **30**.

[0035] The printing unit **30d** includes actuators and various devices, sensors, drive circuits, mechanical parts, etc., for executing printing on a print medium. The sensors include sensors that detect various detection targets that can vary in the printing apparatus **30**. Examples include a sensor that detects the remaining amount of print medium and a sensor that detects the remaining amount of each color material used for printing. In the present embodiment, the printing unit **30d** includes a mechanism for recording color materials of predetermined colors on a print medium. In

the present embodiment, the printing unit **30d** records on the print medium using CMYKGGyROr (C: cyan, M: magenta, Y: yellow, K: black, G: green, Gy: gray, R: red, Or: orange) color materials. Hereinafter, it is assumed that the printing apparatus **30** is an ink jet printer that uses ink as the color material, but the printing method is not limited to the ink jet method. The color materials used for recording are not limited to the CMYKGGyROr combination and can be varied as appropriate. [0036] The printing unit **30d** can print on print media of various sizes. That is, the printing unit **30d** includes a storage unit for storing print media of various sizes and can transport and print the stored print media.

[0037] The UI unit **30e** is composed of an input unit such as buttons and a touch panel provided on the housing of the printing apparatus **30** and a display unit for displaying various information. Based on the information displayed on the display unit of the UI unit **30e**, the user can operate the input unit to issue various instructions.

[0038] The processor **30a** of the printing apparatus **30** performs printing based on the print data **10c3** transmitted from the information processing apparatus **10**. That is, when printing is performed, the print data **10c3** is transmitted along with a print instruction from the information processing apparatus **10** and stored in the non-volatile memory **30c**. Based on the print data **10c3**, the processor **30a** controls the printing unit **30d**, etc., and executes printing on the print medium stored in the printing unit **30d**.

(1-3) Configuration of Information Processing Apparatus

[0039] FIG. **4** is a block diagram illustrating the configuration of the information processing apparatus **10**. In the present embodiment, the information processing apparatus **10** includes a processor **10a**, a communication unit **10b**, non-volatile memory **10c**, a display unit **10d**, and an input unit **10e**. The processor **10a** includes an unillustrated CPU, ROM, RAM, etc., executes various programs recorded in the non-volatile memory **10c**, and can control each unit of the information processing apparatus **10**, the colorimetric instrument **20**, etc. In other words, the processor **10a** functions as a control unit.

[0040] The communication unit **10b** includes a communication interface for communicating with the colorimetric instrument **20** and the printing apparatus **30** according to various wired or wireless communication protocols. The communication unit **10b** may also include an interface for communicating with various removable memories attached to the information processing apparatus **10**.

[0041] Various data can be stored in the non-volatile memory **10c**. In the present embodiment, colorimetric data **20c1**, a color conversion model **10c1**, image data **10c2**, and print data **10c3** are stored in the non-volatile memory **10c**. The colorimetric data **20c1** is data transmitted from the colorimetric instrument **20** and indicates the color value (HSV value), which is the colorimetric result of the sample color.

[0042] The color conversion model **10c1** is data for color-converting image data and associates colors in a device-independent color space with colors in a device-dependent color space. In the present embodiment, the color conversion model **10c1** includes an input-side model that performs input-side color conversion and an output-side model that performs output-side color conversion.

[0043] The input-side model that performs input-side color conversion is a model that converts RGB gradation values or CMYK gradation values indicating colors in a device-dependent color space and HSV values indicating colors in a device-independent color space into each other. In the present embodiment, a model for converting RGB gradation values or CMYK gradation values to HSV values is machine-learned. A model for converting HSV values to RGB gradation values or CMYK gradation values is also machine-learned based on the training data. These learned models are the input-side models. The form of the input-side model may be various; for example, it may be a neural network that converts input values to output values, or it may be composed of a machine-learned model and an optimization module that performs optimization, similar to the output-side model described later. It is also possible to use an existing ICC profile instead of the input-side

model in the color conversion model **10c1**. This ICC profile is a lookup table used for various devices; for example, in the present embodiment, a display profile corresponding to a display used as the display unit **10d** can be used.

[0044] The output-side model that performs output-side color conversion is a model that converts colors in a device-dependent color space, i.e., CMYKGGyROr gradation values indicating the color materials used by the printing apparatus **30**, and HSV values indicating colors in a device-independent color space into each other. In the present embodiment, the output-side model includes a model for converting the usage amounts of the color materials to color values (hereinafter referred to as a color prediction model) and an optimization module.

[0045] Here, the color prediction model includes a model for converting CMYKGGyROr gradation values indicating the usage amounts of the color materials into spectral reflectance. This model is generated, for example, by preparing training data in which CMYKGGyROr gradation values and spectral reflectance are associated in advance and machine-learning a neural network that converts CMYKGGyROr gradation values into spectral reflectance based on the training data. The spectral reflectance can be specified, for example, by measuring a color printed based on the CMYKGGyROr gradation values using the colorimetric instrument **20**. The spectral reflectance can be converted into a color value by using a known calculation method, and the color prediction model also includes this conversion. Therefore, the color prediction model is a model that converts CMYKGGyROr gradation values into color values via spectral reflectance. The optimization module is used to convert a color

[0046] value to be converted into usage amounts of color materials using the color prediction model. Specifically, the optimization module converts a color value to be converted into provisional usage amounts of color materials, inputs them to the color prediction model, and converts them into a color value. The color value obtained by this conversion is called a predicted color value. If the color difference between the predicted color value and the color value to be converted is less than or equal to a reference value, the optimization module considers that the two match. In this case, the color value to be converted is converted into the provisional usage amounts of the color materials.

[0047] On the other hand, if the color difference between the predicted color value and the color value to be converted is greater than the reference value, the optimization module corrects the provisional usage amounts of the color materials. Furthermore, the optimization module inputs the corrected provisional usage amounts of the color materials to the color prediction model and determines whether or not the color difference between the obtained predicted color value and the color value to be converted is less than or equal to the reference value. The optimization module repeats the above process until the color difference between the predicted color value and the color value to be converted becomes less than or equal to the reference value. When the color difference between the predicted color value and the color value to be converted becomes less than or equal to the reference value, the optimization module considers that the color value to be converted has been converted into the provisional usage amounts of the color materials. As described above, according to the output-side model of the color conversion model **10c1**, CMYKGGyROr gradation values can be converted into HSV values by the color prediction model. Also, HSV values can be converted into CMYKGGyROr gradation values by the color prediction model and the optimization module.

[0048] Further, in the present embodiment, constraint conditions for restricting the color materials used in the printing apparatus **30** can be imposed during conversion by the output-side model of the color conversion model **10c1**. Specifically, when converting a color value to be converted into usage amounts of color materials, the optimization module sets provisional usage amounts of the color materials. At this time, it can be set so as not to use specific color materials. For example, when at least one of the CMYKGGyROr color materials is not used, the usage amount of that color material is fixed at 0. With this configuration, the optimization module specifies provisional usage

amounts of the color materials such that the color difference between the predicted color value and the color value to be converted is less than or equal to the reference value while the usage amount of the color material is fixed at 0. Therefore, when provisional usage amounts of the color materials are specified such that the color difference between the predicted color value and the color value to be converted is less than or equal to the reference value, a color value can be converted into usage amounts of the color materials under the constraint of not using specific color materials. When converting usage amounts of color materials into a color value, the usage amounts of the unused color materials may be set to 0 and input to the prediction model.

[0049] The configuration in which color conversion is performed by the color conversion model **10c1** is merely an example, and color conversion may naturally be performed by various functions or lookup tables. However, by using the color conversion model **10c1**, the resources required for the information processing apparatus **10** and the man-hours required for data generation can be reduced. For example, a lookup table is data in which HSV values and colors in a device-dependent color space are associated for multiple representative points, and the number of representative points is generally very large. In particular, when associating CMYKGGyROr gradation values with HSV values, the CMYKGGyROr gradation values are 8-dimensional coordinate values, and it is necessary to associate a large number of points in the 8-dimensional coordinate space with 3-dimensional HSV space coordinate values, resulting in a very large amount of data. In the case of a lookup table, in order to restrict color materials, it is necessary to create a lookup table for all cases after the restriction. For example, when restricting cyan or magenta, it is necessary to create a lookup table for each case.

[0050] On the other hand, although a large amount of data may be required when creating a learned model obtained by machine learning, the amount of data is generally less than that of a lookup table. Further, in the present embodiment, it is not necessary to create different learned models for each condition in order to restrict color materials. Therefore, assuming a situation where there are multiple conditions such that specific color materials are not used and a color conversion model **10c1** and a lookup table corresponding to each condition are prepared, the data amount can be reduced when preparing the color conversion model **10c1** compared to when preparing the lookup table.

[0051] The task of creating a lookup table generally requires a very large number of man-hours. The task of preparing lookup tables for each condition requires an extremely large number of man-hours. However, to generate the output-side model of the color conversion model **10c1** as in the present embodiment, it is not necessary to generate different models for each condition. Therefore, the color conversion model **10c1** can be generated with a smaller configuration than creating a lookup table.

[0052] The image data **10c2** is data indicating an image to be printed. As described above, the image data **10c2** includes image data of multiple plates. In the present embodiment, it is assumed that there is at least one process color plate and one spot color plate. Therefore, the image data **10c2** includes data of the process color plate in which the color of each pixel is specified by an RGB gradation value or a CMYK gradation value. The image data **10c2** also includes data of the spot color plate, including an RGB gradation value or a CMYK gradation value of a representative color specified in the header and a gradation value for each pixel indicating a relative density difference from the representative color. Naturally, this configuration is merely an example, and the number of plates that can exist is not limited, and multiple plates may exist or a specific plate may not exist. The image data **10c2** includes information indicating a weighting factor for superimposing each plate, and an image to be printed is specified by superimposing the data of each plate with the weighting factor.

[0053] The print data **10c3** is data for causing the printing apparatus **30** to execute printing. In the present embodiment, the processor **10a** performs image processing including color conversion by the color conversion model **10c1** based on the image data **10c2** to generate the print data **10c3**.

Specifically, the image data **10c2** includes a printer description language, and the processor **10a** performs rendering processing including analysis and conversion processing into raster data based on the printer description language. After rasterization, the processor **10a** performs color conversion of the raster data using the color conversion model **10c1** and acquires the usage amount of each color material necessary to print the color of each pixel. For example, in the process color plate, the processor **10a** converts the RGB value or CMYK value of each pixel into an HSV value by the input-side model or the ICC profile and converts the HSV value of each pixel into a CMYKGGyROr gradation value by the output-side model. In the spot color plate, the processor **10a** acquires the CMYKGGyROr gradation value of each pixel based on the CMYKGGyROr gradation value of the representative color and the gradation value of each pixel. The processor **10a** performs page layout determination processing, halftone processing, etc. on the print medium to generate the print data **10c3**. When the print data **10c3** is generated, the print data **10c3** is transmitted to the printing apparatus **30**, and printing is performed.

[0054] The display unit **10d** is a display device that displays arbitrary images. The input unit **10e** is a device on which the user performs input operations. The information processing apparatus **10** can be implemented in various ways and may be a stationary computer or a portable computer. In the former case, the display unit **10d** can be configured, for example, with a display independent from the computer main body, and the input unit **10e** can be configured, for example, with a keyboard, mouse, etc., independent from the computer main body. In the latter case, the display unit **10d** and the input unit **10e** can be configured, for example, with a touch panel display integrated with the computer main body. In any case, the user can input the user's intention by operating the input unit **10e** while visually recognizing images and characters displayed on the display unit **10d**. Hereinafter, in the present embodiment, the description will be given assuming that the information processing apparatus **10** is a stationary computer.

[0055] The processor **10a** can execute an unillustrated information processing program. The information processing program according to the present embodiment displays a screen for setting when printing a spot color plate so as to reproduce the sample color measured by the colorimetric instrument **20** on the display unit **10d**. The user can use this screen to print a color chart including patches of the reference color and surrounding colors, and the user can specify the representative color of the spot color plate by selecting a patch on the color chart.

[0056] In the present embodiment, the information processing program has a function for performing color matching using the color chart. When the information processing program is executed, the processor **10a** functions as a color value acquisition unit **10a1**, a reproducibility acquisition unit **10a2**, and a generation unit **10a3**.

[0057] The color value acquisition unit **10a1** has a function of acquiring color values indicating the colors of multiple color patches included in the color chart. Specifically, the color value acquisition unit **10a1** receives information for specifying the color value of each of the multiple color patches from the user. In the present embodiment, the processor **10a** acquires the color value (HSV value) of the sample color measured by the colorimetric instrument **20** and converts the color value into the usage amount of each color material by the output-side model of the color conversion model **10c1**. The HSV value is a value of the device-independent color space.

[0058] The user can correct the usage amount of each color material or restrict the color materials to be used by inputting to a screen, which will be described later; the user can specify the desired usage amounts of the desired color materials. The color specified in this way is the reference color. When the reference color is specified, the user specifies a parameter to be varied among the parameters that specify the color. The parameter to be varied is called a variation parameter. The variation parameter may be various parameters, and in the present embodiment, the user can select three types of variation parameters from among the hue value (H value), the saturation value (S value), the brightness value (V value), and the usage amount of the color materials.

[0059] The user further specifies a variation range of values in the variation parameter. The color

value acquisition unit **10a1** generates surrounding colors by varying the value of the variation parameter among the values of the parameters of the reference color according to the variation range. The number of surrounding colors generated according to the variation range may be fixed or may be specified by the user. When the usage amount of the color materials is included in the variation parameters, the color value acquisition unit **10a1** converts the usage amount into a color value based on the output-side model of the color conversion model **10c1**. According to the above configuration, the user can generate surrounding colors by varying the reference color in a desired manner.

[0060] When the color values of the reference color and surrounding colors are specified, each color is considered to be a color value indicating the color of multiple color patches. That is, according to the above processing, the color values of multiple color patches are acquired. When any two of the acquired multiple color patches are extracted, one can be considered as the first color patch, and the other can be considered as the second color patch. Therefore, it can be said that the color value acquisition unit **10a1** has a function of acquiring a first color value indicating the color of the first color patch included in the color chart and a second color value indicating the color of the second color patch included in the color chart.

[0061] The reproducibility acquisition unit **10a2** has a function of acquiring the color reproducibility when each of the multiple color patches is printed by the printing apparatus **30**. In the present embodiment, the reproducibility is the color difference between the color value indicating the color of the color patch and the printed color value, which is the color value of the color printed by the printing apparatus **30** based on the color value. That is, the color value indicating the color of the color patch indicates the ideal color when printing the color patch. When printing is performed by the printing apparatus **30** based on the color value, the color value is converted into the usage amount of each color material by the output-side model of the color conversion model **10c1**, and the color patch is printed with the usage amount.

[0062] If the conversion by the output-side model of the color conversion model **10c1** is accurate, the color of the printed color patch matches the color value indicating the color of the color patch. However, if the conversion is inaccurate, the color values do not match. Such a situation can occur when the color indicated by the color value cannot be reproduced by the printing apparatus **30**, that is, when the color indicated by the color value is outside the color gamut.

[0063] Therefore, in the present embodiment, the reproducibility acquisition unit **10a2** acquires the printed color value, which is the color value of the color printed by the printing apparatus **30**, and acquires the color difference between the printed color value and the color value indicating the color of the color patch as the reproducibility. The details of these conversions will be described later. As described above, the reproducibility acquisition unit **10a2** acquires the color reproducibility for each color patch, but when any two of the multiple color patches are extracted, one can be considered as the first color patch, and the other can be considered as the second color patch. Therefore, it can be said that the reproducibility acquisition unit **10a2** has a function of acquiring a first reproducibility indicating the color reproducibility when the first color patch is printed by the printing apparatus **30** based on the first color value and acquiring a second reproducibility indicating the color reproducibility when the second color patch is printed by the printing apparatus **30** based on the second color value.

[0064] Furthermore, it can be said that the reproducibility acquisition unit **10a2** has a function of acquiring, as the first reproducibility, the color difference between the first color value and the first printed color value, which is the color value of the color printed by the printing apparatus **30** based on the first color value, and acquiring, as the second reproducibility, the color difference between the second color value and the second printed color value, which is the color value of the color printed by the printing apparatus **30** based on the second color value.

[0065] The generation unit **10a3** has a function of generating print data for printing the color chart. That is, the generation unit **10a3** generates the print data **10c3** for printing the color chart including

the reference color and surrounding colors and stores it in the non-volatile memory **10c**. However, at this time, the generation unit **10a3** alters the representation of the color patch according to the reproducibility. Specifically, when the first reproducibility is lower than the second reproducibility, the generation unit **10a3** generates the print data for printing the color chart in which the first color patch and the second color patch have different representations.

[0066] In the present embodiment, when the color difference indicating the color reproducibility is greater than or equal to the threshold value, the reproducibility is considered low. The threshold value is a value set in advance to evaluate low color reproducibility. In the present embodiment, a color patch for which the color difference indicating the color reproducibility is greater than or equal to the threshold value is the first color patch, and a color patch for which the color difference is smaller than the threshold value is the second color patch. That is, in the present embodiment, whether or not the first reproducibility, which is the color reproducibility of the first color patch, is lower than the second reproducibility, which is the color reproducibility of the second color patch, is determined based on whether or not the color difference is greater than or equal to the threshold value.

[0067] The generation unit **10a3** generates the print data **10c3** so that the first color patch and the second color patch printed on the color chart can be distinguished from each other. In the present embodiment, the first color patch is surrounded by a dashed frame for warning that the color reproducibility is low. The second color patch is not surrounded by a dashed frame. In the present embodiment, the print data **10c3** is generated so that a character string indicating the identification information of the color patch is printed in association with each color patch. Specific examples of such color patches will be described later.

[0068] When the user issues a print instruction while the print data **10c3** is generated, the processor **10a** transmits the print data **10c3** to the printing apparatus **30** via the communication unit **10b**. The printing apparatus **30** receives the print data **10c3** via the communication unit **30b** and executes printing based on the print data **10c3**. As a result, the color chart including the reference color and surrounding colors is printed. The user compares the color patches of the color chart, specifies a desired color patch, and specifies the identification information by the input unit **10e**.

[0069] The processor **10a** specifies the color patch based on the identification information specified by the user, specifies the color value and the usage amount of each color material of the color patch, and acquires it as the representative color of the spot color plate. That is, the color value or the usage amount of each color material is recorded in the header indicating the representative color of the spot color plate.

[0070] As a result, the user's settings are reflected in subsequent printing of the spot color plate. Specifically, when printing the spot color plate, the representative color is printed in the color of the color patch specified by the user. Colors other than the representative color are printed as colors with the density of the color patch specified by the user increased or decreased based on the gradation value associated with each pixel. According to the above configuration, when the user selects a desired color patch from the color chart, the user can recognize the first color patch printed as a color different from the color value when the variation parameter is varied as instructed by the user and the second color patch in which the color value when the variation parameter is varied as instructed by the user is reproduced. Therefore, the user can select the representative color while recognizing whether or not the color accurately reproduces the color value.

(2) Print Control Process

[0071] Next, the color chart print control process will be described in detail with reference to the flowchart shown in FIG. 5. The user prepares the image data **10c2** to be printed and stores it in the non-volatile memory **10c** before starting the print control process. When starting printing, the user operates the input unit **10e** of the information processing apparatus **10** to cause the processor **10a** to execute the print control program.

[0072] When the print control program starts, the processor **10a** of the information processing

apparatus **10** controls the display unit **10d** to display a usage amount specification screen (step **S100**). The usage amount specification screen is a screen for specifying the color value of the reference color and the usage amount of each color material used when the reference color is printed by the printing apparatus **30**.

[0073] FIG. **6** illustrates an example of the usage amount specification screen. On this screen, a character string indicating that it is a screen for creating a color chart is displayed at the top, and the screen can be switched by two tabs shown below it. There are tabs for setting the reference color and for specifying color creation conditions. The screen shown in FIG. **6** illustrates a state where the tab for setting the reference color is selected, and this screen is the default display screen of the usage amount specification screen.

[0074] In the example illustrated in FIG. **6**, a character string indicating the applicable spot color plate is displayed on the left side of the screen, and information on the reference color is displayed on the right side of the screen. On the left side of the screen, the spot color plates included in the image data **10c2** are listed, and the selected spot color plate is colored gray. On the right side of the screen, an icon **I** simulating the reference color is displayed at the top, and a user interface for specifying the usage amount of each color material is displayed at the bottom. The icon **I** simulating the reference color is a rectangular sample simulating the color corresponding to the color value indicating the reference color and is not displayed on the initial screen.

[0075] In the user interface for specifying the usage amount of each color material, the names of the color materials usable in the printing apparatus **30** are displayed in a list, and checkboxes for specifying whether or not to use each color material are associated with the names of the color materials. The color materials specified as color materials to be used by the checkboxes are referred to as use color materials. Icons simulating the colors of the color materials and input boxes indicating the usage amounts of the color materials are associated with the names of the color materials. The usage amount of each color material can be specified as any value from 0 to 100%, but no numerical value is displayed on the initial screen. In the present embodiment, the % value indicating the usage amount of each color material and the gradation value are associated with each other in advance. Naturally, the usage amount of each color material may be specified by a gradation value.

[0076] The user can specify the color values in various ways to specify the reference color, but in this example, the color values is acquired by using the colorimetric result by the colorimetric instrument **20**. Therefore, the user operates the colorimetric instrument **20** to measure the sample color. When the colorimetric measurement is performed, the processor **20a** of the colorimetric instrument **20** acquires the colorimetric data **20c1** (step **S200**) and stores it in the non-volatile memory **20c**.

[0077] Next, the processor **20a** of the colorimetric instrument **20** transmits the colorimetric data **20c1** to the information processing apparatus **10** via the communication unit **20b** (step **S205**). By the function of the color value acquisition unit **10a1**, the processor **10a** of the information processing apparatus **10** receives the colorimetric data **20c1** via the communication unit **10b** (step **S105**) and stores it in the non-volatile memory **10c**. Furthermore, the processor **10a** refers to the input-side model or the display profile of the color conversion model **10c1** and converts the HSV value indicated by the colorimetric data **20c1** into an RGB gradation value. The processor **10a** then controls the display unit **10d** (e.g., display) and displays the icon **I** with the RGB gradation value.

[0078] Next, the color value acquisition unit **10a1** acquires the usage amount of each use color material (step **S110**). The user determines the use color materials, operates the input unit **10e**, and checks the checkboxes displayed on the right side of the usage amount specification screen illustrated in FIG. **6**. According to this configuration, the user can specify all or part of the color materials usable in the printing apparatus **30** as use color materials. The color value acquisition unit **10a1** acquires the checked color materials as use color materials. The unchecked color materials are not acquired as use color materials, and their use is prohibited when printing the color patches.

That is, the presence or absence of checks in the checkboxes defines a constraint condition of using the use color materials and not using the color materials that are not use color materials.

[0079] Furthermore, the color value acquisition unit **10a1** refers to the output-side model of the color conversion model **10c1** and converts the color value acquired in step **S105** into the usage amount of each use color material under the constraint condition of using the use color materials and not using the color materials that are not use color materials. The color value acquisition unit **10a1** controls the display unit **10d** and displays the usage amounts obtained by the conversion in the input boxes indicating the usage amounts of the color materials. Note that the user may be correctable the usage amounts in the input boxes by operating the input unit **10e**. When the usage amounts are corrected by the user, the color value acquisition unit **10a1** refers to the output-side model of the color conversion model **10c1**, converts the usage amounts of each color material into a color value, and considers it as the color value of the reference color.

[0080] Next, the user operates the input unit **10e** and selects the tab for specifying the color creation conditions illustrated in FIG. 6. In response to the selection, the reproducibility acquisition unit **10a2** displays the creation condition specification screen (step **S115**). FIG. 7 illustrates an example of the creation condition specification screen. In the example illustrated in FIG. 7, a diagram schematically illustrating the color chart is displayed on the left side of the screen, and a user interface for specifying creation conditions when varying the reference color to generate surrounding colors is displayed on the right side of the screen.

[0081] In the example illustrated in FIG. 7, the user interface is a screen for setting each of the three variation parameters. The letters X, Y, and Z are respectively associated with the three variation parameters. A type, a variation range, and a number are associated with each variation parameter, and the user can specify the content of each parameter. The type is the type of variation parameter, and by selecting the type, the user can specify which parameter of the color representing the reference color should be varied. In the present embodiment, the user can specify any three of the usage amounts of color materials, the hue value, the saturation value, and the brightness value as the variation parameters. When the variation parameter is the usage amount of the color materials, the color of the color material to be varied can be selected. In the example illustrated in FIG. 7, magenta is selected.

[0082] The variation range is the amount of one step when varying the value of the variation parameter step by step. For example, the user can specify the variation range by a percentage of the usage amount, a hue value, or the like. The number indicates the number of color patches generated by varying the variation parameter. The user can specify the number using numerical values or a slide bar.

[0083] When the variation range and the number are specified, the configuration is determined in which the color patches are arranged. Specifically, the variation parameter associated with X is a parameter that varies when the position of the color patch varies in the horizontal direction, and the variation parameter associated with Y is a parameter that varies when the position of the color patch varies in the vertical direction. These X and Y correspond to the horizontal and vertical axes, and the color patches generated by varying the variation parameter associated with X are arranged horizontally by the number specified for the variation parameter. The color patches generated by varying the variation parameter associated with Y are arranged vertically by the number specified for the variation parameter.

[0084] The color patches arranged vertically and horizontally by the specified numbers as described above are called a block. For example, when the numbers are specified as 3 for both the variation parameter associated with X and the variation parameter associated with Y, 3 color patches are arranged vertically and horizontally in one block. In the example illustrated in FIG. 7, the block located in the center is surrounded by a solid line and displayed with the letter Z attached. The variation parameter associated with Z is a parameter that varies in different blocks. That is, the color patches including multiple blocks printed on the color chart may be configured to differ from

each other in the value of the variation parameter associated with Z.

[0085] When such color patches are compared with each other, the variation parameters associated with X, Y, and Z each vary by the variation range. For example, the colors of horizontally adjacent color patches have a color difference of the variation range in the value of the variation parameter associated with X. The colors of vertically adjacent color patches have a color difference of the variation range in the value of the variation parameter associated with Y. Furthermore, by varying the value of the variation parameter associated with Z by the variation range, colors included in different blocks are generated.

[0086] FIG. 8 illustrates an arrangement example of color patches generated when the settings for X, Y, and Z are the example illustrated in FIG. 7. In this example, since the numbers for X and Y are 3 and 3, respectively, the number of color patches arranged in one block is 9. In the example illustrated in FIG. 8, numerical values (01 to 81), which serve as identification information, are printed in association with each color patch.

[0087] In the color patches illustrated in FIG. 8, the color patch with the identification information 41 arranged in the center is the reference color patch. The block B1 including the reference color is composed of 9 color patches. In the horizontal direction, the usage amount of magenta, which is the variation parameter associated with X, varies by the variation range ΔX . Therefore, for example, the color patch with the identification information 42 is printed with a magenta usage amount that is greater than that of the reference color by ΔX , and the color patch with the identification information 40 is printed with a magenta usage amount that is less than that of the reference color by ΔX .

[0088] In the vertical direction, the brightness value, which is the variation parameter associated with Y, varies by the variation range ΔY . Therefore, for example, the color patch with the identification information 37 is printed with a usage amount corresponding to a color value with a brightness value greater than that of the color patch with the identification information 40 by ΔY , and the color patch with the identification information 43 is printed with a usage amount corresponding to a color value with a brightness value less than that of the color patch with the identification information 40 by ΔY .

[0089] Further, in the color patches illustrated in FIG. 8, 9 blocks are arranged, and in each block, the saturation value, which is the variation parameter associated with Z, varies by the variation range ΔZ . Therefore, for example, the color patch with the identification information 72 in the block B2 is printed with a usage amount corresponding to a color value with a saturation value greater than that of the color patch with the identification information 45 in the block B1 by ΔZ , and the color patch with the identification information 16 in the block B3 is printed with a usage amount corresponding to a color value with a saturation value less than that of the color patch with the identification information 43 in the block B1 by ΔZ . The dashed-dotted lines indicating the blocks, character strings B1, B2, B3, ΔX , ΔY , ΔZ , and arrows in FIG. 8 are for explanation and are not printed on the color chart.

[0090] In the color chart as described above, a warning regarding color reproducibility is printed. Specifically, a dashed frame is added and printed around color patches for which the color difference indicating the color reproducibility, i.e., the color difference between the color value indicating the color of the color patch and the printed color value, which is the color value of the color patch printed by the printing apparatus 30, is greater than or equal to the threshold value. In the example illustrated in FIG. 8, the color patches with identification information 01 to 05, 07, 22, and 25 have a dashed frame indicating low color reproducibility.

[0091] Returning to FIG. 5, the description will be continued. To generate the print data 10c3 for the color chart as described above, the color value acquisition unit 10a1 sets one of the color patches as a target patch and performs the processing of steps S125 to S160 on the target patch, based on the input content of the creation condition specification screen. Specifically, the color value acquisition unit 10a1 sets the target patch (step S125). The target patch is one of the color

patches printed on the color chart and is a color patch that has not been subjected to the loop processing of steps S125 to S160.

[0092] The target patch is set by, for example, the following processing. The color value acquisition unit **10a1** acquires the total number of color patches printed on the color chart based on the number specified for each of the three variation parameters corresponding to X, Y, and Z. The color value acquisition unit **10a1** then associates identification information with each of these color patches. Furthermore, the color value acquisition unit **10a1** sorts the color patches based on the identification information and sequentially selects identification information that has not been subjected to the loop processing of steps S125 to S160 and sets it as the target patch.

[0093] Next, the color value acquisition unit **10a1** acquires the color value of the target patch (step S130). That is, the color value acquisition unit **10a1** acquires the color value and usage amount of each color material acquired in steps S100, S105, and S110 for the reference color. Furthermore, the color value acquisition unit **10a1** specifies the variation range of the variation parameter to be varied from the reference color and the number of variations with the variation range, based on the variation range specified in step S120. Here, the first color value and the second color value are examples of color values acquired in step S130.

[0094] For example, in the example illustrated in FIGS. 7 and 8, when the target patch is the color patch with the identification information **45**, the color value acquisition unit **10a1** specifies the usage amount when the usage amount of magenta of the reference color is increased once by the variation range ΔX . Furthermore, the color value acquisition unit **10a1** refers to the output-side model of the color conversion model **10c1** and converts the obtained usage amounts of each color material into a color value. Furthermore, the color value acquisition unit **10a1** decreases the obtained color value once by the variation range ΔY and considers it as the color value of the target color patch with the identification information **45**. When the target patch is the color patch with the identification information **72**, for example, the color value acquisition unit **10a1** increases the saturation value of the color patch with the identification information **45** once by the variation range ΔZ and considers it as the color value of the target color patch with the identification information **72**.

[0095] When the color value of the target patch is acquired, the reproducibility acquisition unit **10a2** converts the color value into usage amounts of color materials (step S135). That is, the reproducibility acquisition unit **10a2** refers to the output-side model of the color conversion model **10c1** and converts the color value into usage amounts of color materials. Since the value after being converted by the output-side model of the color conversion model **10c1** indicates the usage amounts of the color materials, it is within the color gamut of the printing apparatus **30**. However, the color value, which is the input value to the output-side model of the color conversion model **10c1**, is not necessarily a color within the color gamut of the printing apparatus **30**. In particular, in the present embodiment, even if the reference color is within the color gamut, the surrounding colors may be outside the color gamut by varying the variation parameter.

[0096] Even if the color value of the target patch is outside the color gamut of the printing apparatus **30**, the usage amount of each color material converted by the output-side model of the color conversion model **10c1** is within the color gamut of the printing apparatus **30**. That is, the output-side model of the color conversion model **10c1** according to the present embodiment may convert a color value outside the color gamut of the printing apparatus **30** into a color within the color gamut of the printing apparatus **30**. FIG. 9 is a diagram illustrating color values in a device-independent color space. In FIG. 9, the inside of the boundary B is an area within the color gamut of the printing apparatus **30**. FIG. 9 also shows the color gamut of the printing apparatus **30** on a surface obtained by cutting the device-independent color space in a direction perpendicular to a certain brightness axis. In this example, it is assumed that C1 is the color value of the reference color. When surrounding colors are generated from the reference color by varying the values of the variation parameter, the color value C2 of the surrounding color may be outside the color gamut.

[0097] Even with such a color value C2, when conversion is performed by the output-side model of the color conversion model **10c1**, it is converted into usage amounts of the color materials. The color printed with the usage amounts in the printing apparatus **30** is a color within the color gamut of the printing apparatus **30**. In FIG. 9, the color value of the color within the color gamut is indicated as color value C3. As described above, in the present embodiment, colors outside the color gamut can be converted into colors within the color gamut. When such conversion is performed, the accuracy of color conversion by the output-side model of the color conversion model **10c1** is degraded, and the color reproducibility is degraded.

[0098] Therefore, in the present embodiment, the reproducibility acquisition unit **10a2** re-converts the usage amounts of the color materials obtained by the conversion in step S135 into color values (step S140). That is, the reproducibility acquisition unit **10a2** refers to the output-side model of the color conversion model **10c1** and converts the usage amounts of the color materials obtained in step S135 into color values.

[0099] The reproducibility acquisition unit **10a2** then acquires the color difference between the color value acquired in step S130 and the color value acquired in step S140 and determines whether or not the color difference is greater than or equal to the threshold value (step S145). That is, the reproducibility acquisition unit **10a2** considers that the color reproducibility is lower than the reference when the color difference is greater than or equal to the threshold value, and considers that the color reproducibility is within the reference range when the color difference is smaller than the threshold value. Here, the first reproducibility is the reproducibility corresponding to the first color value, and is an example of reproducibility where the color difference is less than the threshold and within the reference range. The second reproducibility is the reproducibility corresponding to the second color value, and is an example of reproducibility where the color difference is greater than or equal to the threshold and outside the reference range.

[0100] When it is determined in step S145 that the color difference is greater than or equal to the threshold value, the generation unit **10a3** generates a patch including a warning (step S150). Specifically, the generation unit **10a3** generates image data for printing a uniform rectangular color patch of a predetermined size using each color material with the usage amount acquired in step S135 and printing a dashed frame around the color patch. Here, the second color patch corresponding to the second color value and the second reproducibility is an example of a color patch with a warning, that is, a color patch with a dashed frame printed around it.

[0101] On the other hand, if it is not determined in step S145 that the color difference is greater than or equal to the threshold value, the generation unit **10a3** generates a patch without a warning (step S155). Specifically, the generation unit **10a3** generates image data for printing a uniform rectangular color patch of a predetermined size using each color material with the usage amount acquired in step S135. Here, the first color patch corresponding to the first color value and the first reproducibility is an example of a color patch without a warning, that is, a color patch with no dashed frame printed around it.

[0102] When step S150 or step S155 is executed, the generation unit **10a3** determines whether or not all the patches have been processed (step S160). That is, the generation unit **10a3** determines that all patches have been processed when the processing of steps S125 to S160 has been performed on all color patches printed on the color chart as target patches. If it is determined in step S160 that all patches have not been processed, the processor **10a** repeats the processing from step S125 onward.

[0103] On the other hand, if it is determined in step S160 that all the patches have been processed, the generation unit **10a3** generates the print data **10c3** of the color chart (step S165). Specifically, the generation unit **10a3** arranges the color patch images generated in steps S150 and S155 in the order illustrated in FIG. 8. That is, the generation unit **10a3** arranges the color patches so that when the position varies to the next position in the horizontal direction, the variation parameter corresponding to X varies by the variation range, and when the position varies to the next position

in the vertical direction, the variation parameter corresponding to Y varies by the variation range. In this way, blocks are formed. The generation unit **10a3** arranges the blocks so that the variation parameter corresponding to Z varies by the variation range between adjacent blocks. Furthermore, the generation unit **10a3** generates the print data **10c3** so that the usage amount of each color material in each color patch is the usage amount acquired in step **S135**.

[0104] Next, the generation unit **10a3** determines whether or not a print instruction has been issued (step **S170**). Specifically, when the user operates the input unit **10e** and issues an instruction using the print button **Bp** illustrated in FIGS. **6** and **7**, the generation unit **10a3** determines that a print instruction has been issued. If it is determined in step **S170** that a print instruction has been issued, the generation unit **10a3** outputs the print data **10c3** (step **S175**). That is, the generation unit **10a3** outputs the print data **10c3** to the printing apparatus **30** via the communication unit **10b**.

[0105] The processor **30a** of the printing apparatus **30** acquires the print data **10c3** via the communication unit **30b** and prints the color chart (step **S300**). The user compares the color patches printed on the color chart, selects a desired color patch, operates the input unit **10e**, and inputs the identification information of the selected color patch. The processor **10a** considers that the representative color is specified by the color value and usage amounts of color materials of the color patch corresponding to the input identification information. According to the above configuration, the user's desired color can be set as the representative color of the spot color plate.

[0106] In the above configuration, when the user compares the color patches printed on the color chart, the user can know whether or not the color is different from the color value generated by varying the variation parameter as instructed, based on the presence or absence of the dashed frame. That is, the user can know whether or not the color reproducibility is low. Furthermore, in the present embodiment, even when the color reproducibility is low, a configuration is adopted in which the color patch is printed with a warning rather than not being printed. Therefore, even if the color reproducibility is low, the user can select the color patch with the low reproducibility if it is closest to the desired color. Therefore, compared to when the color patch with low color reproducibility is not printed, the choices available to the user can be increased.

(3) Other Embodiments, Etc.

[0107] The above embodiments are merely examples for carrying out the present disclosure, and various other embodiments can be adopted. For example, the information processing apparatus **10** and other apparatuses (at least one of the colorimetric instrument **20** and the printing apparatus **30**) may be provided integrally. The connection mode of each apparatus is not limited to the configuration illustrated in FIG. **1**, and communication between arbitrary apparatuses may be performed via an arbitrary network. Furthermore, at least some of the functions of each apparatus may be distributed among multiple apparatuses; for example, at least some of the functions of the information processing apparatus **10** may be implemented by a cloud server. At least some of the functions of the information processing apparatus **10** may be implemented by other apparatuses to form an information processing system.

[0108] The color value acquisition unit only needs to be able to acquire a first color value indicating the color of the first color patch included in the color chart and a second color value indicating the color of the second color patch included in the color chart. That is, the color value acquisition unit acquires color values indicating multiple colors. The method for acquiring the color values may be various methods. Therefore, various configurations may be adopted in addition to the configuration in which the color of the color sample is measured for the reference color. For example, color values may be associated with identification numbers of color samples, and the color values may be acquired based on the identification numbers. Alternatively, color values may be specified by the user. The first color value and the second color value may be any different color values. Since the number of patches included in the color chart is not limited to two, one of any two of the patches included in the color chart may be the first color value and the other may be the second color value, respectively. The color value may be any value that indicates a color, and

spectral reflectance may be a color value, for example, in addition to coordinate values in a color space.

[0109] The reproducibility acquisition unit only needs to be able to acquire a first reproducibility indicating the color reproducibility when the first color patch is printed by the printing apparatus based on the first color value, and acquire a second reproducibility indicating the color reproducibility when the second color patch is printed by the printing apparatus based on the second color value. That is, the degree to which the printing apparatus can reproduce the color of any color value when printing it varies depending on the color. Therefore, the reproducibility acquisition unit only needs to be able to acquire the color reproducibility in the first color patch and the second color patch. The reproducibility may be any index for evaluating the degree to which the color of the printed color patch matches the color indicated by the color value. The aspect of the reproducibility is not limited and may be indicated either by the color difference or whether or not it is within the color gamut.

[0110] The generation unit only needs to be able to generate print data for printing the color chart. That is, the generation unit only needs to be able to generate print data for printing a color chart including the first color patch and the second color patch. However, when the first reproducibility is lower than the second reproducibility, the generation unit generates print data for printing a color chart in which the first color patch and the second color patch have different representations. That is, the color chart is printed so that the user can recognize the color patch for which the reproducibility is relatively low. The number of patches included in the color chart is not limited to two, and may be three or more.

[0111] The method to allow the user recognize that the first reproducibility of the first color patch is lower than the second reproducibility is not limited to the method using the dashed frame as described above. Specifically, the first color patch and the second color patch printed on the color chart may be configured to differ from each other in at least one of the pattern printed around the patch, the character printed around the patch, the patch shape, the patch size, and the patch position.

[0112] The form of the pattern printed around the patch is not limited to the dashed frame as described above, and may be a pattern formed by various figures, colors, etc. The configuration may also be such that the user recognizes that the first reproducibility is lower than the second reproducibility by the characters printed around the patch. Furthermore, the configuration may also be such that the user recognizes that the first reproducibility is lower than the second reproducibility because of different patch shapes, sizes, and positions. Furthermore, these various methods may be combined. Furthermore, an explanatory sentence indicating that a dashed frame or the like is attached to the color patch with low color reproducibility may be printed on the color chart.

[0113] The reproducibility may be any index for evaluating the difference between the color indicated by the color value and the color value of the color actually printed when the color value is printed and is not limited to the configuration indicated by the color difference. For example, the configuration may also be such that the processor **10a** acquires the color gamut of the printing apparatus **30** and acquires the relationship between the first color value and the color gamut as the first reproducibility and acquires the relationship between the second color value and the color gamut as the second reproducibility.

[0114] The color gamut may be indicated by, for example, multiple color values existing on the boundary of the color gamut or multiple color values existing within the color gamut. When the color gamut is acquired, the processor **10a** can define the reproducibility based on the relationship between the color value and the color gamut. For example, when the color value exists outside the color gamut, it can be considered that the color reproducibility when the color patch is printed by the printing apparatus **30** based on the color value is lower than when the color value exists within the color gamut.

[0115] Therefore, if the color reproducibility when the color value is outside the color gamut is assumed to be the first reproducibility and the color reproducibility when the color value is within the color gamut is assumed to be the second reproducibility, and the representations of the first color patch and the second color patch are made different, the user can specify the color patch for which the reproducibility is low.

[0116] The color conversion is not limited to the configuration in which it is performed by the color conversion model **10c1**. That is, in the configuration in which the first color value and the second color value are converted into usage amounts of color materials based on any conversion data that converts a color value into usage amounts of color materials to generate print data, a configuration with different color patch representations is applicable. For example, even in a configuration in which color conversion is performed using a color conversion profile, a lookup table, a combination thereof, or the like, the accuracy of the color conversion may be lowered for colors near the boundary of the color gamut, colors included in areas where the data density of the lookup table is lower than in other parts, etc. Therefore, in order to allow the user to recognize the first color patch with relatively low reproducibility, the first color patch may be printed in a representation different from the second color patch. According to the above configuration, the user can be made aware of deterioration in color reproducibility in configurations using various conversion data.

[0117] Furthermore, in the color chart, it may be possible to select whether or not to print the first color patch with low color reproducibility. For example, the generation unit generates the print data for printing the color chart including the first color patch in the first mode in which the first color patch is printed on the color chart when the first reproducibility is lower than a threshold value, and generates the print data for printing the color chart not including the first color patch in the second mode in which the first color patch is not printed on the color chart when the first reproducibility is lower than the threshold value.

[0118] Such a configuration is implemented, for example, by the configuration in which the user selects the first mode or the second mode before step **S100** or before or after any of steps **S100** to **S120** in the above-described embodiment. In the first mode, the processing from step **S125** onward is executed similarly to the processing illustrated in FIG. 5. In the second mode, step **S150** is omitted in the processing from step **S125** onward. As a result, image data is not generated for the target patch determined in step **S145** to have a color difference greater than or equal to the threshold, and in step **S165**, print data not including the target patch is generated. According to this configuration, by having the user who does not want to select a color with low reproducibility select the second mode, erroneous selection of a color with low reproducibility can be prevented.

[0119] Furthermore, the above-described systems, programs, and methods may be implemented as a single apparatus or by using parts included in multiple apparatuses, and include various aspects. Further, they can be modified as appropriate, such as a part being software and a part being hardware. Furthermore, the disclosure also holds as a recording medium for a program for controlling an apparatus or system. Naturally, the recording medium for the program may be a magnetic recording medium, a semiconductor memory, or any recording medium developed in the future.

Claims

1. An information processing apparatus comprising: a color value acquisition unit that acquires a first color value indicating a color of a first color patch included in a color chart and a second color value indicating a color of a second color patch included in the color chart; a reproducibility acquisition unit that acquires a first reproducibility indicating color reproducibility when the first color patch is printed by a printing apparatus based on the first color value and acquires a second reproducibility indicating color reproducibility when the second color patch is printed by the

- printing apparatus based on the second color value; and a generation unit that generates print data for printing the color chart, wherein when the first reproducibility is lower than the second reproducibility, the generation unit generates the print data for printing the color chart in which the first color patch and the second color patch have different representations.
2. The information processing apparatus according to claim 1, wherein the reproducibility acquisition unit acquires, as the first reproducibility, a color difference between the first color value and a first printed color value, which is a color value of a color printed by the printing apparatus based on the first color value, and acquires, as the second reproducibility, a color difference between the second color value and a second printed color value, which is a color value of a color printed by the printing apparatus based on the second color value.
 3. The information processing apparatus according to claim 1, wherein the reproducibility acquisition unit acquires a color gamut of the printing apparatus, acquires a relationship between the first color value and the color gamut as the first reproducibility, and acquires a relationship between the second color value and the color gamut as the second reproducibility.
 4. The information processing apparatus according to claim 1, wherein the first color patch and the second color patch printed on the color chart differs from each other in at least one of a pattern printed around the patch, a character printed around the patch, a patch shape, a patch size, and a patch position.
 5. The information processing apparatus according to claim 1, wherein the color value acquisition unit acquires a variation parameter, which is a parameter to be varied among parameters representing color values, and a variation range of values of the variation parameter, and acquires the first color value and the second color value by varying the value of the variation parameter by the variation range.
 6. The information processing apparatus according to claim 1, wherein the first color value and the second color value are values in a device-independent color space, and the generation unit converts the first color value and the second color value into usage amounts of color materials, based on conversion data that converts color values into usage amounts of color materials, to generate the print data.
 7. The information processing apparatus according to claim 1, wherein the color value acquisition unit acquires use color materials and usage amounts of the use color materials to be used by the printing apparatus and acquires the first color value and the second color value based on the usage amounts of the use color materials.
 8. The information processing apparatus according to claim 1, wherein the generation unit generates the print data for printing the color chart including the first color patch in a first mode in which the first color patch is printed on the color chart when the first reproducibility is lower than a threshold value, and generates the print data for printing the color chart not including the first color patch in a second mode in which the first color patch is not printed on the color chart when the first reproducibility is lower than the threshold value.
 9. A non-transitory computer-readable storage medium storing an information processing program, the program causing a computer to function as: a color value acquisition unit that acquires a first color value indicating a color of a first color patch included in a color chart and a second color value indicating a color of a second color patch included in the color chart; a reproducibility acquisition unit that acquires a first reproducibility indicating color reproducibility when the first color patch is printed by a printing apparatus based on the first color value and acquires a second reproducibility indicating color reproducibility when the second color patch is printed by the printing apparatus based on the second color value; and a generation unit that generates print data for printing the color chart, wherein when the first reproducibility is lower than the second reproducibility, the generation unit generates the print data for printing the color chart in which the first color patch and the second color patch have different representations.
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